NIHR | National Institute for Health and Care Research

Reference number	NIHR207190
Programme	Research for Patient Benefit
Call	RfPB Competition 52 Stage 2
Lead Applicant	Dr Karen Tiley
Host organisation	Oxford University Hospitals NHS Foundation Trust
Research Title	Assessing normal ranges of antibody levels in healthy children at different ages, for clinical reference purposes
Start Date	01/11/2024
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Research Type	Secondary Research

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1. The Research Team

Specify your (lead applicant) role in the research
Karen Tiley will be responsible for the collation of data sources, co-ordination of laboratory testing, data analysis, statistical methodology, preparation of results, reporting and dissemination.

% FTE Commitment: This refers to the percentage of your time that you will commit to this project. If you are funded as part of other NIHR projects that will be running concurrently, your time must not exceed 100% overall.
40

Co-applicant role and % FTE commitment
Specify the role and percent FTE commitment of each co-applicant.

Full Name	Dr Dominic Kelly	**Institution**	University of Oxford

Please indicate if this co-applicant is a member of the public
No

Specify role in research
Clinical lead, involved in interpretation of results and chart design, drawing on his experience in the paediatric and immunology team in the Oxford Children's Hospital.

% FTE Commitment: This refers to the percentage of your time that you will commit to this project.
2

Full Name	Ms Sophia Wilkinson	**Institution**	University of Oxford

Please indicate if this co-applicant is a member of the public
No

Specify role in research
The patient and public involvement (PPI) lead will implement the PPI plan outlined in this proposal, to ensure that there is consistent lay involvement over the life-cycle of this research. The lead will coordinate the activities of a small PPI group, providing contributors with the necessary practical, training and financial support so their involvement is as beneficial as possible to them, the research and patients.

% FTE Commitment: This refers to the percentage of your time that you will commit to this project.
5

2. History of application

Has this application been previously submitted to this or any other funding body?
No

3. Scientific abstract

Scientific abstract

Background

In a clinical paediatric setting, measurement of serum antibody-responses to routinely administered childhood vaccines (tetanus, Haemophilus Influenzae B (Hib) and pneumococcal) is undertaken to give an indication of a child's immune function.

Recognised levels of antibody, above which a child is considered protected (thresholds of clinical protection) are available for tetanus and Hib, but the pneumococcal test produces 13 antibody results so a decision on whether a child has mounted an adequate immune response is harder to reach.

The pneumococcal vaccine (PCV13) protects against 13 different versions (serotypes) of the pneumococcal bacteria. Currently a single antibody threshold is applied to each of the 13 antibody results so the clinician knows how many results out of 13 are above the clinical threshold of protection. However, this information can be difficult to interpret because even in healthy children with a fully functional immune system antibody levels may be below the threshold of clinical protection for some serotypes. This will increasingly be the case for older children as antibody wanes, rather than indicating any problem with the immune system.

Currently there are no reference data describing normal pneumococcal antibody responses at each age against which individual test results can be compared.

Objectives

This study aims to develop a single reference chart for use in clinical practice that will help paediatricians to interpret pneumococcal antibody test results. The chart will be based on normal ranges of pneumococcal antibody responses in healthy children of different ages. It will be similar in format to the percentile growth charts that are currently used to indicate normal ranges of height and weight in young children.

Methods

The chart will be informed by analysis of pneumococcal antibody data from healthy children across ages 0-10 years of age (children older than 10 years would not have received PCV13). Some of these data have already been measured as part of previous epidemiological studies in young children (0-3 years). Data for ages 4-10 years will be generated by testing stored residual blood samples from a previous seroprevalence study.

To find normal ranges we will calculate percentiles of the distribution of antibodies for each year of age. In addition we will summarise whether it is 'normal' for a child to have antibody above the commonly used threshold value. Antibody wanes quickly after vaccination so older children who are entirely normal are likely to have antibody below the threshold.

In order to translate this analysis into a practical tool, we will work with members of the public and paediatricians to develop a chart that can be easily used in clinical practice and understood by parents.

Timelines

Testing of the stored blood samples will be completed within the first six months of the project. The analysis should be completed within the next eight months, with publication and dissemination shortly afterwards.

Impact and Dissemination

Initially the chart will be published in a national medical journal, additional plans for dissemination into clinical practice will be discussed with paediatricians and the public.

4. Plain English Summary

Plain English summary

Children who have repeated infection are often referred to hospital to check if the child's immune system is working normally. Part of this testing includes measuring how well the child has responded to vaccines they received as an infant. A poor response might indicate that there is a problem with their immune system. These tests usually include responses to tetanus, *Haemophilus influenzae* type B (Hib) and pneumococcal vaccines.

Pneumococcal bacteria can cause serious illnesses like meningitis (infection in the brain and spinal cord), sepsis (life-threatening reaction to an infection) and pneumonia (infection in the lungs).

Healthy children will each produce different amounts of antibody (proteins that recognise germs and alert the immune system, so it can destroy them) when vaccinated. To understand a child's pneumococcal test results, it would be helpful to compare them against antibody data for healthy children (reference data) as is done for the other vaccines. However, these data are not currently available to doctors.

In this study we will generate reference data by studying antibodies to pneumococcal vaccine in healthy children representative of the general population in terms of geography, ethnicity and sex. Most of these data are already available from previous studies that have measured pneumococcal antibody responses for other purposes. Some data will be obtained from testing stored blood samples.

These data will then be used to understand individual test results for children in hospital. We will develop a chart that hospital doctors can use to assess whether a child's pneumococcal antibodies fall within the range of healthy children of a similar age. This is a similar concept to infant and child growth charts.

These reference data will allow a more informed decision about whether the child's immune system has responded to previous vaccines like other healthy children. Currently, doctors may recommend a booster vaccine if they think the child's vaccine response might be low but this judgement is relatively subjective. If the reference data shows the antibody response is in the range of values seen in healthy children, unnecessary booster vaccines can be avoided. This will save general practice resources and reduce parental anxiety. By using the pneumococcal reference chart, hospital doctors will save time that would otherwise be spent asking more specialist colleagues for advice about test results. Most importantly, the chart will give doctors the information they need to decide on the best treatment for the child.

Six patient and public involvement (PPI) contributors, of mixed genders and ethnicities, including several parents, have been consulted about the research design. They will also support the implementation of the research, in particular helping to develop the chart and a plan for its dissemination, convinced of the benefits to clinical practice, and therefore patients.

5. Changes from first stage

NIHR | National Institute for Health and Care Research

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Wednesday 14 February 2024

Dear Dr Tiley,

Research for Patient Benefit (RfPB) Programme: NIHR207190 – Assessing normal ranges of antibody levels in healthy children at different ages, for clinical reference purposes

Following the Stage 1 review of applications submitted to RfPB Competition 52, I am pleased to inform you that your application has been invited to progress to Stage 2 of the application process.

Further to our previous correspondence, I include your feedback from the Committee below. You are advised to address these feedback points in Stage 2 of the application process.

The Committee made the following comments:

- The Committee commented that the proposal was expensive for a study which may require further work in order to achieve patient benefit. Based on the trajectory to patient benefit, this study would be expected to be delivered within the Tier 3 funding limit. Therefore, the team was asked to reduce the costs or provide a strong justification of the proposed costs in the Stage 2 application.

- The rationale of the proposed study was unclear. The research team was requested to:
    - Further clarify the scale of the problem being addressed.
    - Describe the trajectory to patient benefit in more detail.
    - Outline the outcomes from the proposed study and the future development of the research.
    - Clarify how the proposed research would encourage change in clinical practice.

- Concerns were raised regarding some aspects of the research plan and methodology, and the applicants were invited to address them in the Stage 2 application. Specifically, the applicants were asked to:
    - Clarify how representative the dataset is of ‘normal’ ranges and if all the different age ranges would be covered sufficiently.
    - Justify the requirement for new data to be collected (in addition to the existing datasets that are planned to be analysed).
    - Provide more details in the analysis plan.

    - Consider involving someone else in the analysis work to avoid introducing bias when interpreting the results.

- The Committee recommended patient and public involvement in the development of the charts for clinical use.

- The Committee requested the plain English summary to be rewritten to remove acronyms such as PCV13.

The attached outcomes table contains the anonymised mean scores and ranking from the Competition 52 Stage 1 Regional Assessment Committee meeting where your application was assessed. It is intended to aid you in contextualising your feedback from the Committee.

Your anonymised code is: SEC-0124-0006

The Stage 2 application form is now available to complete on the Research Management System and the deadline for submission is **1pm** on **Wednesday 20 March 2024.**.

Please be aware that applications should be accompanied with the 'Funder Export' from the online SoECAT, obtainable via the NIHR Central Portfolio Management System (CPMS). Further guidance is available via the RMS application.

Further information on the RfPB Programme, including guidance notes and frequently asked questions, can be found on the RfPB website. The RfPB Programme also encourages applicants to share the Committee's feedback and speak with the NIHR Research Support Service (RSS) for further advice.

I wish you all the best for Stage 2 of the Competition.

Yours sincerely,

Mr Jack Finn
Programme Manager

Changes from Stage 1

The Committee commented that the proposal was expensive for a study which may require further work in order to achieve patient benefit. Based on the trajectory to patient benefit, this study would be expected to be delivered within the Tier 3 funding limit. Therefore, the team was asked to reduce the costs or provide a strong justification of the proposed costs in the Stage 2 application.

We have reduced the total project costs by reducing the amount of laboratory testing, whilst still incorporating additional PPI costs.

Considering the availability of multiple data sources for younger age groups (Tables 1 and 2), we have decided to reduce testing of the STORY samples to focus on ages three to ten years inclusive. We are retaining age three years for validation against existing data sources. We have also reduced the number of samples tested in each age group (40 compared with 100) as this should still provide sufficient numbers for calculation of mean titres and percentiles (to an accuracy of 2.5 compared with 1 percentile).

The laboratory testing of the STORY samples is an important component of the project because these samples were collected as part of a seroprevalence study from a nationally representative cohort of children. The laboratory undertaking the testing is a WHO reference laboratory that has undertaken testing for several of the studies in younger age groups; this will provide us with high quality, consistent results.

The rationale of the proposed study was unclear. The research team was requested to:

- **Further clarify the scale of the problem being addressed.**

General paediatricians in hospital practice across the UK are frequently asked to see children who appear to have more infections than would be expected for a child of their age. One aim of the paediatric review is to decide whether a child has a primary immunodeficiency (rare genetically determined conditions which predispose children to unusual, frequent or severe infections). The number of children who are assessed for infection is much higher than those who turn out to have a primary immunodeficiency. Therefore careful clinical and laboratory assessment are needed to differentiate healthy children from those who need more detailed genetic investigation. An important part of the laboratory assessment includes measuring the ability to make antibodies to vaccines and **most of the children** being assessed for recurrent infection would have antibody levels measured. These include antibodies to vaccines for *Haemophilus influenzae* type b (Hib), tetanus and pneumococcal vaccines as these vaccines are given to all children and laboratory tests for these vaccine responses are widely available.

For many of the assessments of immunity there are clearly normal ranges (e.g. total immunoglobulins, lymphocyte numbers) but for vaccine responses – which have a substantial degree of variation between children and also decrease after vaccination – the normal ranges are not clearly defined. Instead antibody levels have been defined as putative thresholds of protection – for Hib and tetanus there are single values as there is a single vaccine antigen. For pneumococcus the situation is more complex; there are 13 values as the vaccine contains 13 different types of antigen to protect against 13 of the most common pneumococcal bacteria. It is not expected that every child will have a level of antibody above the putative protective threshold for each of the 13 pneumococcal serotypes. It is well recognised that there is sustained protection in a population of children immunised with the UK schedule even when antibodies have decreased for some serotypes with time after vaccination below the putative protective threshold. This is likely to be due to the degree of herd immunity in a population with good vaccine coverage. There are no reference data easily available to a clinician that describes how many serotypes (and which serotypes) would be above the putative threshold of protection in a healthy child of any specific age who would not be considered at risk of pneumococcal infection. Instead the clinician is faced with making a subjective judgement about the number of serotypes with a low response that should trigger a booster dose being offered, or raising a suspicion that there is an immunological problem or discussing each result with a specialist – even where the probability of an immunodeficiency is thought to be low from the overall clinical assessment.

Quantifying the scale of this problem: one of the co-applicants works in a tertiary hospital in both general paediatrics and paediatric infectious disease and immunology and the scale and nature of the clinical practice would be similar to many other tertiary hospitals in the UK. Vaccine responses are measured on around 2-3 clinic patients weekly (100-150 times / year) in the infectious disease clinic (where children have been referred for a specialist opinion) and questions about interpreting vaccine responses are frequent from other general paediatricians in the tertiary hospital and surrounding district general hospitals **who are seeing a much larger number of these patients**. The interpretation of vaccine responses – particularly to the pneumococcal vaccine - is one of the most frequent immunological questions asked of the specialty team by general paediatricians. A previous assessment showed that the immunology laboratory in the tertiary hospital had measured pneumococcal antibody responses on >900 samples annually prior to the COVID pandemic – these data include adult and children's samples and some repeat

measurements but demonstrates that this is a widely used assay which would benefit from clearer interpretative criteria.

- **Describe the trajectory to patient benefit in more detail.**

The reference data on pneumococcal vaccine responses will be developed into a tool or reference chart that can be readily interpreted by the clinicians requesting the tests. This chart would be of relevance to and made accessible for all general paediatricians in the UK and to specialist paediatricians in infection and immunology clinics.

Reference data relevant to healthy children will provide clarity in interpretation of test results so that clinicians can make more informed decisions about the immune function of a child and therefore what further course of treatment may or may not be needed.

Children may avoid unnecessary referral for additional vaccination if their results are within the range expected of typical healthy children

It is also possible that in the absence of clear reference data, some clinicians may not be recommending vaccination that would be advisable to protect the child from pneumococcal infection following consultation of the chart.

By providing a clear way for clinicians to interpret these test results, we will save time that would otherwise be spent referring results to more specialist colleagues so the main focus can be on the next steps for the patient.

- **Outline the outcomes from the proposed study and the future development of the research.**

The main outcome for this study is a clinical tool in the form of a chart that brings together the information needed to clearly interpret children's pneumococcal antibody results at different ages.

The methodology around development of the chart will be published and could potentially be applied to other areas.

Once developed, further work could include developing clinical training around use of the chart, and assessing the impact of the chart in clinical practice.

In the future the chart could be adapted to respond to changes in the vaccine or schedule used within the UK as and when new data become available.

In the future we could also explore whether this chart could be used in other countries that have introduced PCV13, or if the methodology of the chart could be shared internationally.

- **Clarify how the proposed research would encourage change in clinical practice**

If the chart is available to clinicians it will reduce the ambiguity currently associated with pneumococcal test results and provide a clear and consistent process for interpreting these vaccine responses. It should also reduce the time taken to refer results to specialist colleagues so that more time is available to focus on patient care.

Concerns were raised regarding some aspects of the research plan and methodology, and the applicants were invited to address them in the Stage 2 application. Specifically, the applicants were asked to:

- **Clarify how representative the dataset is of 'normal' ranges and if all the different age ranges would be covered sufficiently.**

The children participating in the carriage studies and clinical trials were healthy, without underlying health issues. Importantly we also have the pneumococcal vaccine history for these children so we can use these data to give an indication of the range of responses to pneumococcal vaccine.

The data from the Oxford University Hospital database will contain a wider range of children with and without underlying health issues however, children with diagnosed immunodeficiencies will be identified and excluded from the main analysis.

The STORY samples were collected as part of a seroprevalence study specifically designed to sample children representative of the general population.

The data spans the full range of ages of children who have received PCV13 (ages 0-10 years inclusive). We have added a table into the plan to show this more clearly (Table 2).

- **Justify the requirement for new data to be collected (in addition to the existing datasets that are**

planned to be analysed).

The new data involves laboratory testing residual stored samples that were collected as part of a separate study (STORY). These data are needed to cover the ages not available from other studies (4 to 10 years). Whilst these ages are also available within the hospital database, the hospital population will usually be children referred because of a clinical concern about infections or immunity as so, even where a primary immunodeficiency was not identified, will not necessarily be representative of completely healthy children. The STORY dataset is representative of the general population which is important for capturing normal ranges.

- **Provide more details in the analysis plan.**

We have added more detail in the analysis plan about data validation and development of methodology.

- **Consider involving someone else in the analysis work to avoid introducing bias when interpreting the results.**

The analysis will be verified by a member of the statistics team at the Oxford Vaccine Group as part of our internal validation process.

The analysis will also be presented at Statistics Team meetings within the Oxford Vaccine Group at each stage to enable critique and methodological suggestions to be incorporated.

We will also discuss the methodology and interpretation with colleagues from other organisations that specialise in vaccines and immunology, for example the UK Health Security Agency (UKHSA) and colleagues at our collaborator organisation (University College London).

The interpretation of results will also be discussed with a group of both general paediatricians and specialists in paediatric infectious disease and immunology in order to guide how best to present the reference data to facilitate decision making for use in clinic.

The Committee recommended patient and public involvement in the development of the charts for clinical use.

We have incorporated this suggestion into the revised plan and are proposing to hold a workshop to support the development of the charts, to ensure that it can also be used as an explanation aid, should a clinician wish to share the chart with the parent or carer of their paediatric patient to support their conversation.

The Committee requested the plain English summary to be rewritten to remove acronyms such as PCV13.

We have removed acronyms from the plain English summary. It has been re-reviewed by four lay people and further feedback from this review has also been incorporated.

6. Detailed Research plan

Using all of the headings in the order presented below, please use this section to clearly explain your proposed research. Schematics, tables, illustrations, graphs, and other types of graphics can be embedded to clarify the research plan but they should not clutter the central narrative. Images do not count towards the overall word count but inclusion of them to overcome word limits is not permitted. Images may only be included within the 'Research Plan.' Images included in other sections will be removed from the application and not seen by reviewers.

For further information for what is required under each heading please refer to the Guidance for Applicants. **As this is the main part of your application which will be considered by the reviewing committee, you should ensure that the information is accurate, succinct, clearly laid out and provides sufficient methodological detail.**

NOTE: Applicants should aim to reserve a significant proportion of the word count for the project plan to enable methodological approaches to be fully specified.

1. Background and Rationale
2. Aims and Objectives
3. Research Plan / Methods
4. Dissemination, Outputs and anticipated Impact
5. Project / research timetable

6. Project Management
7. Ethics / Regulatory Approvals
8. Project / Research expertise
9. Success criteria and barriers to proposed work

1. Background and Rationale

What is the problem being addressed?

General paediatricians in hospital practice across the UK are frequently asked to see children who appear to have more infections than would be expected for a child of their age. One aim of the paediatric review is to decide whether a child has a primary immunodeficiency (rare genetically determined conditions which predispose children to unusual, frequent or severe infections). The number of children who are assessed for infection is much higher than those who turn out to have a primary immunodeficiency. Therefore careful clinical and laboratory assessment are needed to differentiate healthy children from those who need more detailed genetic investigation. An important part of the laboratory assessment includes measuring the ability to make antibodies to vaccines and most of the children being assessed for recurrent infection would have vaccine antibody levels measured. These include antibodies to *Haemophilus influenzae* type b (Hib), tetanus and pneumococcal vaccines as these vaccines are given to all children and laboratory tests for these vaccine responses are widely available.

For many of the other laboratory assessments of immunity there are clear normal ranges (e.g. total immunoglobulins, lymphocyte numbers); but for vaccine responses – which have a substantial degree of variation between children and also decrease after vaccination – normal ranges are not defined. Instead antibody levels are interpreted with reference to putative thresholds of protection. For Hib and tetanus the vaccine responses are single values as there is a single vaccine antigen for which the quantity of antibody is measured. For pneumococcus the situation is more complex. Pneumococcal conjugate vaccine is included in the UK routine childhood immunisation schedule to provide protection against pneumococcal disease, which in its severest forms can cause pneumonia, meningitis and septicaemia. There are over 100 different versions (serotypes) of the pneumococcal bacteria (*Streptococcus pneumoniae*). The current vaccine (PCV13) protects against 13 different serotypes and induces 13 separate antibody responses which are measures in the laboratory. It is not expected that every child will have a level of antibody above the putative protective threshold for each of the 13 pneumococcal serotypes as antibody levels decrease with time following immunisation.

A standard cut off of 0.35 IU/mL is typically used as a general threshold to determine whether the antibody against any specific serotype is likely to protect against infection [1]. Confusion arises when some serotypes fall above the threshold and some below as will happen even in completely healthy children who would not be considered at risk of pneumococcal infection. This may be completely normal for a particular age, given the different strengths of response to each serotype and waning immunity, but these data are not currently readily available to clinicians who need to interpret test results. Instead the clinician is faced with making a subjective judgement about what number of serotypes with a low response should trigger a booster dose being offered, or raise a suspicion that there is an immunological problem or require a discussion with a specialist – even where the probability of an immunodeficiency is thought to be low from the overall clinical assessment.

PCV13 has been part of the routine childhood immunisation schedule in the UK since April 2010, but in January 2020 the number of doses given changed from 2, 4 and 12 months (2+1) to 3 and 12 months of age (1+1) [2] . The impact of this change in terms of normal levels of antibody at each age is also unknown.

Why is this research important in terms of improving the health and/or wellbeing of the public and/or to patients and health care services?

Understanding the results of the antibody test is important for several reasons:

- First and foremost, correct interpretation of pneumococcal antibody levels will appropriately guide further treatment action for the child. With the correct data for comparison clinicians can understand not just whether or not a child has mounted a normal antibody response, but also how good or poor their response is relative to the response in a healthy child.
- Currently, if a clinician is unsure of the interpretation, to err on the side of caution, booster vaccinations may be recommended which causes unnecessary pain for the child, anxiety for the parents, and wastes general practice time and resources.
- It is also possible that in the absence of clear reference data, some clinicians may not be recommending vaccination that would be advisable following consultation of reference data from healthy children.
- Finally, clinicians often refer unclear results to more specialist paediatric teams which takes up consultant time that could be better spent elsewhere.

The main outcome for this study is a clinical tool in the form of reference data presented in an accessible format (for example, a centile chart) that brings together the information needed to clearly interpret children's pneumococcal antibody results at different ages. If the data are available to clinicians it will reduce the subjective interpretation of responses currently associated with pneumococcal test results and provide a clear and consistent process for interpreting these vaccine responses.

Review of existing evidence - How does the existing literature support this proposal?

Defining normal ranges for pneumococcal IgG antibody is problematic, as normal ranges are different for different serotypes and also vary with age [3,4].

Mean IgG titres and the proportions above a particular threshold can be seen to vary for each PCV13 serotype; this has been observed following the primary immunisation series in the Netherlands with significant variation in antibody kinetics between serotypes [5], and post-booster within a healthy population of UK children aged 13-48 months [6]. The PCV13 booster is scheduled at 12 months of age, and data from clinical trials have shown increases in IgG titres in children at this age [7], followed by substantial waning [8], but estimates are not available beyond this for the general population.

In 2003, prior to PCV13 introduction, one study attempted to develop an algorithm for interpreting clinical data on pneumococcal antibody levels which included incorporating serotype-specific invasive pneumococcal disease incidence into an assessment of risk [9]. However, normal ranges across age groups were not defined and the vaccines considered in the study are no longer licensed.

A recent US study described serotype-specific cut offs for normal ranges in healthy adults in relation to the adult pneumococcal polysaccharide vaccine (PPV23) responses and highlighted the need for additional studies in healthy children [10].

Our proposal, of an approach that looks across all serotypes with reference to normal ranges in healthy children in order to help decision-making in clinical practice, is not something that we have found evidence of previously in the literature.

2. Aims and Objectives

The main aim of this study is to develop a clinical reference tool in the form of a single chart that brings together the information needed to clearly interpret children's pneumococcal antibody test results at different ages.

From the chart clinicians will be able to see that for example at 7 years of age if a child has antibody responses above the threshold of clinical protection for 10 out of 13 serotypes, they are above the 50th percentile for their age and therefore that immune response would be typical of a healthy child with the same schedule of previous pneumococcal immunisation.

To best understand normal ranges of pneumococcal vaccine responses in young children in order to inform development of the chart, we will interrogate our data with respect to the following questions:

1. What are normal ranges for pneumococcal serotype-specific IgG titres in healthy UK children at different ages?

a. How many serotype-specific IgG titres fall below 0.35 IU/mL (recognised general pneumococcal IgG threshold of protection) at each year of age?

b. How many serotype-specific IgG titres fall below other published serotype-specific correlates of protection [1] at each year of age?

c. Which serotype titres are maintained above recognised thresholds at each age?

d. What is the distribution (e.g. 5%, 50%, 95% percentiles) of serotype-specific titres at each age?

2. What is the extent of variability within the data sources?

3. How does the change in schedule from 2+1 to 1+1 affect the normal ranges?

3. Research Plan / Methods

This study would involve the compilation of data from studies and trials which have measured pneumococcal serotype-specific IgG in young age groups (0-3 years), and the laboratory assessment of serotype-specific IgG in stored blood samples from a recent UK seroprevalence study (3-10 years, inclusion of age three years in both datasets enables validation).

The children participating in the carriage studies and clinical trials to be used for this work were healthy, without underlying health issues. Importantly we also have the pneumococcal vaccine history for these children so we can use these data to give an indication of the range of responses to pneumococcal vaccine.

The data spans the full range of ages of children who have received PCV13 (ages 0-10 years inclusive).

The data sources and analysis are described below.

Data sources

Pneumococcal serotype-specific IgG results or stored serum samples from healthy children of different ages have been identified from studies undertaken in the UK post PCV13 introduction (Tables 1 and 2). For younger children pneumococcal serotype-specific IgG data are available for both the 2+1 and 1+1 schedules for comparison.

Table 1. Data/samples collected from children aged 0-10 years in England

Schedule	Clinicaltrials.gov or ISRCTN identifier	Data collection time period/ date of randomisation (trials)	Age of children	Number of children with serology results	Analysis laboratory	Data availability
2+1	NCT01996007	2014/15	13-48 months	741	RIVM	Data already available
	NCT03102840	2017/19	13-48 months	498	RIVM	Data already available
	NCT03102840	2017/20	6-12 months	249	RIVM	Data already available
	NCT02482636	2015/16	13 months	91	UCL	Data already available
	NCT02482636	2015/16	21-33 months	35	UCL	Data already available
	NCT04061382	2019/21	3-10 years	-	[UCL]	Samples available
	[hospital database]	2016 - 2022	All ages	>1000	OUHFT	To be shared by OUHFT
1+1	NCT02482636	2015/16	13 months	86	UCL	Data already available
	NCT02482636	2015/16	21-33 months	41	UCL	Data already available
	ISRCTN85819697	2019/20	5 months	174	UCL	Data already available
	ISRCTN85819697	2019/20	13 months	164	UCL	Data already available

Abbreviations: RIVM (National Institute for Public Health and the Environment, The Netherlands), UCL (University College London), OUHFT (Oxford University Hospitals Foundation Trust)

Table 2. Age group availability across the study datasets

		Age (years)										
Schedule	Clinicaltrials.gov or ISRCTN identifier	0	1	2	3	4	5	6	7	8	9	10
2+1	NCT01996007											
	NCT03102840											
	NCT03102840											
	NCT02482636											
	NCT02482636											
	NCT04061382											
	[hospital database]											
1+1	NCT02482636											
	NCT02482636											
	ISRCTN85819697											
	ISRCTN85819697											

Pneumococcal carriage studies

- **NCT01996007 and NCT03102840:** these pneumococcal carriage studies were undertaken by the Oxford Vaccine Group to assess nasopharyngeal carriage and serology in young children [6,11].

Clinical trials

- **NCT02482636:** this trial and follow-up compared the two PCV13 schedules (2+1 vs 1+1) prior to implementation [7,8]. We have received these data from Professor David Goldblatt (University College London, study collaborator) for use in this study.
- **ISRCT85819697**: children participating in both arms of this trial comparing two hexavalent (DTaP-IPV-Hib-HepB) vaccines received PCV13 according to the 1+1 schedule [12].

Oxford University Hospitals Foundation Trust database

Data from the hospital database, concerning children who have had blood samples tested for pneumococcal IgG for clinical assessment purposes, have been extracted for analysis. Children with immunodeficiencies will be identified and excluded from the main analysis. These data will be anonymised prior to use.

National serosurvey: NCT04061382

This national seroprevalence survey, “What’s the STORY?” (Serum Testing of Representative Youngsters), collected sera from children and young people aged 0-24 years old between October 2019 and June 2021 to assess the feasibility of establishing a national serosurvey in England [13]. Residual sera from this study (>100 from each yearly age band) are biobanked with ethical approval for future research. Children and young adults included within “What’s the STORY?” were born between 1995 and 2021. Children born in the UK between April 2010 and early 2020 received PCV13 at 2, 4 and 12 months of age (2+1 schedule); children born between these dates and aged 3-10 years inclusive at the time of blood collection would be the focus of this study (40 from each one-year age band, i.e. 320 samples). Older children within “What’s the STORY?” (aged 11+ years) would have received a lower valency vaccine (PCV7) or were born prior to PCV7 introduction. Testing for the purpose of this study falls under the Oxford Vaccine Centre BioBank protocol [14]. Testing would require 250µl per sample and would be carried out at the WHO reference laboratory for pneumococcal serology, Great Ormond Street Institute of Child Health, University College London, London, UK.

The STORY samples were collected as part of a seroprevalence study specifically designed to sample children representative of the general population. The additional laboratory testing is needed to cover the ages not available from other studies (4 to 10 years, plus age 3 years for validation against other data sources). Whilst these ages are available within the hospital database, this is a specific population of children who have been tested due to illness. The STORY dataset is representative of the general population which is important for capturing normal ranges. Vaccination histories were collected as part of this seroprevalence survey so we can select children who were vaccinated according to the UK schedule.

<u>Data analysis</u>

This analysis described below will form the knowledge base for development of the chart.

The analyses will be verified by a member of the statistics team at the Oxford Vaccine Group as part of our internal validation process. The analysis will also be presented at Statistics Team meetings within the Oxford Vaccine Group

at each stage to enable critique and methodological suggestions to be incorporated.

We will further discuss the methodology and interpretation with colleagues from other organisations that specialise in vaccines and immunology, for example the UK Health Security Agency (UKHSA) and colleagues at our collaborator organisation (University College London).

1. What are normal ranges for pneumococcal serotype-specific IgG titres in healthy UK children at different ages?

a. How many serotype-specific IgG titres fall below 0.35 IU/mL (recognised general pneumococcal IgG threshold) at each year of age?

This would be presented as, for example, a centile chart similar to the infant growth charts, so the number of serotypes above the threshold value could be put in context for the child's age. Methodology for developing appropriate centile charts would be explored.

b. How many serotype-specific IgG titres fall below published serotype-specific correlates of protection at each year of age?

Whilst 0.35 IU/mL is the standard threshold, each serotype has specific published correlates of protection [1] which may be more appropriate, and therefore would be explored in a similar chart.

c. Which serotype titres are maintained above recognised thresholds at each age?

This analysis recognises the variation in waning of different serotype-specific titres and can be used in conjunction with (a) and (b) for interpretation. These data would be presented as a survival-style plot. This analysis would explore both the 0.35 IU/mL threshold and serotype-specific correlates of protection.

d. What is the distribution of serotype-specific titres at each age?

This analysis will examine the distribution (e.g. 5%, 50%, 95% percentiles) of serotype-specific titres at each age. Clinicians could then compare the child's specific titre(s) on these charts to see where the child fits within the range, thus providing reference points independent of threshold-settings.

2. What is the extent of variability within the data sources?

This part of the analysis would explore possible sources of bias such as laboratory, calendar year, sex, ethnicity and study. The data included in this analysis will come from different laboratories, however all use a standardised assay and the World Health Organization reference serum [15,16], so data should be comparable. Data for the same age groups and schedules from different sources will be compared and the extent of variability quantified. Where sufficient data are available, responses will be compared for males and females and different ethnic groups. This information can then be used to adapt the charts to ensure generalisability.

3. How does the change in schedule from 2+1 to 1+1 affect the normal ranges?

Data from children receiving PCV under both the 2+1 and 1+1 schedules are available for the youngest age groups and can be compared to assess the extent of variability with the two different schedules. If variation is minimal, the 2+1 chart across the full age range could reasonably be used by children on either schedule.

<u>Development of charts for clinical use</u>

A group of paediatricians (both general and infectious disease and immunology specialists) and the project's PPI group will be convened to discuss the best approaches for presenting the data for ease of use in clinical practice. Based on their feedback a final chart with guidance notes, incorporating the information from across the different analyses, will be developed for use in clinical paediatric settings.

4. Dissemination, Outputs and Anticipated Impact

The main output for this study is a clinical tool in the form of a chart that brings together the information needed to clearly interpret children's pneumococcal antibody results at different ages.

Using some of the data we have available for children aged 13-48 months, we developed an early template of a possible guidance chart which has already been used to guide clinical decision-making in Oxford. A more comprehensive chart, developed using robust methodology, across a wider age range and based on larger datasets would be a valuable clinical tool that could be implemented across the NHS.

We anticipate that the chart would be used whenever doctors are interpreting pneumococcal test results, and could help when explaining the results to parents. This will streamline and standardise interpretation of these results across the NHS, and hopefully increase patient/parent understanding regarding the reasons for any suggested next steps in terms of treatment.

The final chart will be submitted for publication in a national medical journal, and methodology and results discussed at an international conference such as the ISPPD (International Society of Pneumonia and Pneumococcal Diseases) conference and the European Society for Paediatric Infectious Diseases.

Further work could include developing clinical training around use of the chart, and assessing the impact of the chart in clinical practice.

In the future the chart could be adapted to respond to changes in the vaccine or schedule used within the UK as and when new data become available.

In the future we could also explore whether this chart could be used in other countries that have introduced PCV13, or if the methodology of the chart could be shared internationally.

5. Project / research timetable

Timetable key milestones and deliverables:

- Proposed start date [1 November 2024]
- Extraction, shipping and testing of STORY stored blood samples [1 May 2025]
- Chart development meetings with paediatricians and PPI group [July 2025]
- Finalisation of results and chart [1 January 2026]
- End of study feedback meeting with PPI group [April 2026]

Please see Gantt chart for more detail.

6. Project Management

The project will be managed by Karen Tiley, with reference to the Gantt chart, including regular team meetings with co-investigators to discuss progress. The project is sponsored by the University of Oxford and hosted by Oxford University Hospitals Foundation Trust and progress will also be monitored by research teams within both organisations.

7. Ethics / Regulatory Approvals

This is a secondary analysis of anonymised data so does not require any further ethics approvals.

8. Project / Research expertise

<u>Team</u>

Statistical analysis – Dr Karen Tiley is an experienced Senior Epidemiologist with specific expertise in the analysis and interpretation of large serological datasets, including experience in working with study data relating to pneumococcal carriage and antibody measurements. She is based within the Statistics Team at the Oxford Vaccine Group and will also draw on the advice and experience of colleagues within the team. She previously worked in the

Immunisation Department at the UK Health Security Agency.

Clinical input – Dr Dominic Kelly works at both the Oxford Vaccine Group and the Children's Hospital in Oxford as a BRC Consultant in Paediatrics and Vaccinology within the infection and immunology team. He has extensive research experience with a particular focus on pneumococcal disease in children.

Prof David Goldblatt will oversee the immunological testing of samples from children aged 3-10 years. He is a global expert on pneumococcal vaccines in children and will provide expertise in interpreting study findings.

PPI lead – Sophia Wilkinson works at the Oxford Vaccine Group, as Vaccine Knowledge and Public Engagement Manager, a role that includes responsibility for facilitating PPI. This includes management of OVG's long-standing PPI group of around 20 members. She has an extensive background in health communication, and previously managed several PPI groups for clinical trials at another Medical Science department at the University of Oxford.

8. Success criteria and barriers to proposed work

This project depends on the development of a chart that is useful to clinicians and understood by members of the public. The feedback from these groups will be critical in evaluating the success of the project.

Potential barriers include delays in testing or access to data. The hospital dataset may require more time-consuming data cleaning than anticipated but as this is taking place at the start of the project whilst laboratory testing is ongoing, additional time can be allowed.

Task number	Description	Nov-24	Dec-24	Jan-25	Feb-25	Mar-25	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Apr-26
1	PPI induction and support needs assessment (online)																		
2	Merging of available datasets																		
3	Extraction and cleaning of hospital dataset																		
4	Progress update to PPI group (by email)																		
5	Extraction of STORY samples and shipping to University College London																		
6	Laboratory testing of STORY samples																		
7	Data analysis and development of provisional charts																		
8	Feedback meeting with paediatricians																		
9	Mid-way feedback meeting and reference chart development workshop with PPI group (in person)																		
10	Finalisation of results and chart																		
11	Dissemination (abstract submissions, manuscript writing)																		
12	End of study feedback to PPI group (online)																		

7. Patient & Public Involvement

Please describe how patients and the public have been involved in developing this proposal.

A group of six people, who are members of the Oxford Vaccine Centre's standing PPI group, have been consulted in the development of this research. The group self-selected, based on their interest in the project, and were a mix of genders and ethnicities. It included two parents of non-adult children.

To facilitate discussion Dr Tiley first presented her proposed research, and its rationale, during an online meeting. She specifically sought the group's views with regards to the benefits of the research, and its design. The presentation and discussion format allowed the group to ask questions throughout to ensure effective comprehension.

They concluded that the research undoubtedly would bring benefits to patients. They mentioned that having reference charts could help to reduce parental stress, as previously they may have been told a child had a low immune response, when in fact it was in the normal range. The group also saw that the research could mean children being given fewer unnecessary booster vaccines, which would also save resources, prevent discomfort and remove any vaccine-related risk. An additional benefit the group pointed out was potentially fewer trips to medical appointments.

Several members mentioned the importance of ensuring that the data used to generate the reference charts was

representative of the UK population. In response we adapted the analysis plan to evaluate representativeness and, where possible, separate analyses by ethnicity and sex.

The group also edited the lay summary, giving extensive feedback to simplify and clarify it. Dr Tiley made numerous changes such as removing detail about schedules and explaining where the data would come from.

Finally, the group were consulted about the level of PPI they thought suitable for this research. They concluded that as it is secondary research, with no patient participation, or data collection, it would be sufficient for the group to be consulted at the mid-point and end. This has been expanded slightly, to include an induction and chart development workshop; to ensure as much benefit from PPI as possible, to the patients and the research itself.

Please describe the ways in which patients and the public will be actively involved in the proposed research, including any training and support provided.

a. PPI approach, management and support

A group of five people, who have already been involved in this application, will support the research. From the original six consulted, those with young children will be encouraged to remain involved, because of the relevance of their recent lived experience. As this is secondary research, PPI involvement will be reasonably light-touch.

The PPI lead will manage the group, ensuring meetings and consultations take place at convenient times. The lead will facilitate needs-based support – e.g. training about the research methods – and administer a “needs-assessment”. Group members will be compensated for their involvement.

After an induction, the group will be given short, emailed progress updates every two months before being consulted at a mid-point meeting. This will be an opportunity to discuss any unexpected issues, as well as end of research public engagement and dissemination.

The group will also take part in the development of the reference charts in a workshop. The aim of this is to produce charts that clinicians can use to help explain results to parents in a way that is understandable to them. The group will also be informed of the end results of the research project.

To evaluate the impact of PPI, a framework will be developed to capture feedback at each contact point, along with the change that happened as a result. An end of project survey will also be administered, so the group can share their experiences and inform future PPI practice.

PPI Lead

The PPI Lead has five years’ experience leading public and patient involvement and engagement with the University of Oxford. She has also taught the PPI module of an online MSc in Clinical Trials. She will support Dr Tiley, by organising meetings, taking minutes and providing two-way feedback to the PPI group. She will be responsible for ensuring the group are paid.

b. Summary:

a. Support needs assessment – survey distributed at project start
b. Induction meeting – online due to location of group members
c. Mid-point meeting and reference chart development workshop – in person (Oxford)
d. End of project meeting and evaluation survey - online

In rare cases where proposals do NOT involve patients and the public, clear justification must be provided.

8. Detailed Budget

Note: The SoECAT Form is included as a separate document.

Staff Costs

Budget Item 1

Staff Posts and Salaries

Staff name	Karen Tiley

Staff type	Lead & Co-applicant

Organisation name	University of Oxford

Organisation type	HEI (UK)

Role	Principal Investigator

Grade	08

Salary (£)
48,470

On-costs (£)
12,704

Current annual salary (£)	61,174.00

Annual Costs of Staff Posts

Rate (%) Please note, HEI staff and costs items are funded up to 80%
80

Base annual salary with rate	£48,939.20

	Year 1	**Year 2**	
The year 1 value should always be 0%.	0	2.78	

	Year 1	**Year 2**	
FTE	40	40	

	Year 1	**Year 2**	
Months	12	6	

	Year 1 (£)	Year 2 (£)	Total (£)
Total rate calculation	£19,576.00	£10,060.00	£29,636.00

Budget Item 2

Staff Posts and Salaries

Staff name	Dominic Kelly

Staff type	Lead & Co-applicant

Organisation name	University of Oxford

Organisation type	HEI (UK)

Role	Co-Investigator

Grade	E82

Salary (£)
98,310

On-costs (£)
27,057

Current annual salary (£)	125,367.00

Annual Costs of Staff Posts

Rate (%) Please note, HEI staff and costs items are funded up to 80%
80

Base annual salary with rate	£100,293.60

	Year 1	Year 2	
The year 1 value should always be 0%.	0	1.7	

	Year 1	Year 2	
FTE	2	2	

	Year 1	Year 2	
Months	12	6	

	Year 1 (£)	Year 2 (£)	Total (£)
Total rate calculation	£2,006.00	£1,020.00	£3,026.00

Budget Item 3

Staff Posts and Salaries

Staff name	Sophia Wilkinson

Staff type	PPIEP

Organisation name	University of Oxford

Organisation type	HEI (UK)

Role	VACCINE KNOWLEDGE AND PUBLIC ENGAGEMENT MANAGER

Grade	08

Salary (£)
51,411

On-costs (£)
13,550

Current annual salary (£)	64,961.00

Annual Costs of Staff Posts

Rate (%) Please note, HEI staff and costs items are funded up to 80%
80

Base annual salary with rate	£51,968.80

	Year 1	Year 2	
The year 1 value should always be 0%.	0	2.8	

	Year 1	Year 2	
FTE	5	5	

	Year 1	Year 2	
Months	12	6	

	Year 1 (£)	Year 2 (£)	Total (£)
Total rate calculation	£2,598.00	£1,336.00	£3,934.00

Budget Item 4

Staff Posts and Salaries

Staff name	TBC

Staff type	Research Staff

Organisation name	University of Oxford

Organisation type	HEI (UK)

Role	Biobank Manager

Grade	7

Salary (£)
44,373

On-costs (£)
11,524

Current annual salary (£)	55,897.00

Annual Costs of Staff Posts

Rate (%) Please note, HEI staff and costs items are funded up to 80%
80

Base annual salary with rate	£44,717.60

	Year 1	Year 2	
The year 1 value should always be 0%.	0	2.8	

	Year 1	Year 2	
FTE	5	5	

	Year 1	Year 2	
Months	12	6	

	Year 1 (£)	Year 2 (£)	Total (£)
Total rate calculation	£2,236.00	£1,149.00	£3,385.00

Budget Item 5

Staff Posts and Salaries

Staff name	TBC

Staff type	Research Staff

Organisation name	University of Oxford

Organisation type	HEI (UK)

Role	Statistician

Grade	7

Salary (£)
44,323

On-costs (£)
11,424

Current annual salary (£)	55,747.00

Annual Costs of Staff Posts

Rate (%) Please note, HEI staff and costs items are funded up to 80%
80

Base annual salary with rate	£44,597.60

	Year 1	Year 2	
The year 1 value should always be 0%.	0	0	

	Year 1	Year 2	
FTE	5	0	

	Year 1	Year 2	
Months	8	0	

	Year 1 (£)	Year 2 (£)	Total (£)
Total rate calculation	£1,487.00	£0.00	£1,487.00

Budget Item 6

Staff Posts and Salaries

Staff name	David Goldblatt

Staff type	Lead & Co-applicant

Organisation name	University College London

Organisation type	HEI (UK)

Role	Co-Investigator

Grade	Clinical Academic Consultant MyHR APR2022 Consultant / BAND

Salary (£)
105,701

On-costs (£)
31,971

Current annual salary (£)	137,672.00

Annual Costs of Staff Posts

Rate (%)

Please note, HEI staff and costs items are funded up to 80%
80

Base annual salary with rate	£110,137.60

	Year 1	Year 2	
The year 1 value should always be 0%.	0	0	

	Year 1	Year 2	
FTE	12.5	12.5	

	Year 1	Year 2	
Months	12	5	

	Year 1 (£)	Year 2 (£)	Total (£)
Total rate calculation	£13,767.00	£5,736.00	£19,503.00

Budget Item 7

Staff Posts and Salaries

Staff name	Lucy Roalfe

Staff type	Research Staff

Organisation name	University College London

Organisation type	HEI (UK)

Role	Researcher

Grade	HERA MyHR DEC2023 UCLGR7 / 40

Salary (£)
49,794

On-costs (£)
19,525

Current annual salary (£)	69,319.00

Annual Costs of Staff Posts

Rate (%) Please note, HEI staff and costs items are funded up to 80%
80

Base annual salary with rate	£55,455.20

	Year 1	Year 2	
The year 1 value should always be 0%.	0	0	

	Year 1	Year 2	
FTE	10	10	

	Year 1	Year 2	
Months	12	5	

	Year 1 (£)	Year 2 (£)	Total (£)
Total rate calculation	£5,546.00	£2,311.00	£7,857.00

Budget Item 8

Staff Posts and Salaries

Staff name	Sara Ragab

Staff type	Research Staff

Organisation name	University College London

Organisation type	HEI (UK)

Role	Technician

Grade	HERA MyHR DEC2023 UCLGR6 / 26

Salary (£)
32,982

On-costs (£)
17,152

Current annual salary (£)	50,134.00

Annual Costs of Staff Posts

Rate (%) Please note, HEI staff and costs items are funded up to 80%
80

Base annual salary with rate	£40,107.20

	Year 1	Year 2	
The year 1 value should always be 0%.	0	0	

	Year 1	Year 2	
FTE	15	0	

	Year 1	Year 2	
Months	12	0	

	Year 1 (£)	Year 2 (£)	Total (£)
Total rate calculation	£6,016.00	£0.00	£6,016.00

Travel, Subsistence & Conference Costs

Budget Item 1

Item description	Travel and Subsistence

Organisation Name	University of Oxford

Type of Cost	Travel

Organisation Type	HEI (UK)

Rate %	80

	Year 1	Year 2	
Costs	£933.00	£467.00	

	Year 1 (£)	Year 2 (£)	Total (£)
Total rate calculation	£746.00	£374.00	£1,120.00

Equipment Costs

Consumables

Patient & Public Involvement, Engagement and Participation Costs

Budget Item 1

Item description	PPI

Organisation Name	University of Oxford

Organisation Type	HEI (UK)

Rate %	80

	Year 1	Year 2	
Costs	£1,917.00	£958.00	

	Year 1 (£)	Year 2 (£)	Total (£)
Total rate calculation	£1,534.00	£766.00	£2,300.00

Other Direct Costs

Budget Item 1

Item description	Conferences

Organisation Name	University of Oxford

Organisation Type	HEI (UK)

Rate %	80

	Year 1	Year 2	

Costs	£400.00	£200.00	

	Year 1 (£)	Year 2 (£)	Total (£)
Total rate calculation	£320.00	£160.00	£480.00

Budget Item 2

Item description	Shipping

Organisation Name	University of Oxford

Organisation Type	HEI (UK)

Rate %	80

	Year 1	Year 2	
Costs	£3,333.00	£1,667.00	

	Year 1 (£)	Year 2 (£)	Total (£)
Total rate calculation	£2,666.00	£1,334.00	£4,000.00

Budget Item 3

Item description	Consumables

Organisation Name	University College London

Organisation Type	HEI (UK)

Rate %	80

	Year 1	Year 2	
Costs	£21,139.00	£8,861.00	

	Year 1 (£)	Year 2 (£)	Total (£)
Total rate calculation	£16,911.00	£7,089.00	£24,000.00

Dissemination Costs

Indirect Costs

Budget Item 1

Item description	DA estates

Organisation Name	University of Oxford

Type of Cost	Estate Costs

Organisation Type	HEI (UK)

Rate % Please note for HEIs in the UK, up to 80% of FEC will be paid.
80

	Year 1	**Year 2**	
Costs	£17,095.23	£8,032.71	

	Year 1 (£)	**Year 2 (£)**	**Total (£)**
Total rate calculation	£13,676.00	£6,426.00	£20,102.00

Budget Item 2

Item description	DA Infrastructure Technicians

Organisation Name	University of Oxford

Type of Cost	Estate Costs

Organisation Type	HEI (UK)

Rate % Please note for HEIs in the UK, up to 80% of FEC will be paid.
80

	Year 1	**Year 2**	
Costs	£488.59	£229.59	

	Year 1 (£)	**Year 2 (£)**	**Total (£)**

Total rate calculation	£391.00	£184.00	£575.00

Budget Item 3

Item description	Indirect Costs

Organisation Name	University of Oxford

Type of Cost	Other Indirect Costs

Organisation Type	HEI (UK)

Rate % Please note for HEIs in the UK, up to 80% of FEC will be paid.
80

	Year 1	**Year 2**	
Costs	£42,569.04	£20,002.32	

	Year 1 (£)	**Year 2 (£)**	**Total (£)**
Total rate calculation	£34,055.00	£16,002.00	£50,057.00

Budget Item 4

Item description	DA estates

Organisation Name	University College London

Type of Cost	Estate Costs

Organisation Type	HEI (UK)

Rate % Please note for HEIs in the UK, up to 80% of FEC will be paid.
80

	Year 1	**Year 2**	
Costs	£4,073.00	£1,697.00	

	Year 1 (£)	Year 2 (£)	Total (£)
Total rate calculation	£3,258.00	£1,358.00	£4,616.00

Budget Item 5

Item description	DA Infrastructure Technicians

Organisation Name	University College London

Type of Cost	Estate Costs

Organisation Type	HEI (UK)

Rate % Please note for HEIs in the UK, up to 80% of FEC will be paid.
80

	Year 1	**Year 2**	
Costs	£294.00	£122.00	

	Year 1 (£)	Year 2 (£)	Total (£)
Total rate calculation	£235.00	£98.00	£333.00

Budget Item 6

Item description	Indirect Costs

Organisation Name	University College London

Type of Cost	Other Indirect Costs

Organisation Type	HEI (UK)

Rate % Please note for HEIs in the UK, up to 80% of FEC will be paid.
80

	Year 1	**Year 2**	
Costs	£15,451.00	£6,438.00	

	Year 1 (£)	Year 2 (£)	Total (£)
Total rate calculation	£12,361.00	£5,150.00	£17,511.00

Justification of costs

Vaccine responses are measured on around 2-3 clinic patients weekly (100-150 times / year) in the infectious disease clinic (where children have been referred for a specialist opinion) and questions about interpreting vaccine responses are frequent from other general paediatricians in the tertiary hospital and surrounding district general hospitals who are seeing a much larger number of these patients. The interpretation of vaccine responses – particularly to the pneumococcal vaccine - is one of the most frequent immunological questions asked of the specialty team by general paediatricians. A previous assessment showed that the immunology laboratory in the tertiary hospital had measured pneumococcal antibody responses on >900 samples annually prior to the COVID pandemic – these data include adult and children's samples and some repeat measurements but demonstrates that this is a widely used assay which would benefit from clearer interpretative criteria.

Cost savings that would result from the availability of the chart tool proposed in this project include:
• time saved that would otherwise be spent referring results to more specialist colleagues
• reducing unnecessary referral for additional vaccination and other potential investigations should children be identified as having a response to the pneumococcal vaccine that is within the range of other healthy children of the same age

The proposal relies mainly on existing data sources which are already available or can be accessed free of charge. The laboratory testing of the stored samples is an important component of the project because it provides data for older age groups that are not available from previously conducted studies. These samples were collected as part of a seroprevalence study from a nationally representative cohort of healthy children. Data obtained from testing these samples will be important in itself but also as a means of validating and assessing the representativeness of the other data sources.

The laboratory undertaking the testing is a WHO reference laboratory that has undertaken testing for several of the studies in younger age groups; this will provide us with high quality, consistent results.

We have factored in additional PPI costs, including an in-person chart development workshop which we feel is important to ensure the chart can easily be understood by parents.

There are no NHS Support or Excess Treatment Costs associated with the research. The research is a secondary analysis of previously collected data so no patients will be recruited to the study. There is no cost for extracting the Oxford University Hospitals data. Dr Kelly and Dr Tiley both hold/have applications underway for honorary contracts which will enable them to access the data and prepare it for analysis.

Totals

	Year 1 (£)	Year 2 (£)	Total (£)
Staff Costs	£53,232.00	£21,612.00	£74,844.00
Travel, Subsistence & Conference Costs	£746.00	£374.00	£1,120.00
Equipment Costs	£0.00	£0.00	£0.00
Consumables	£0.00	£0.00	£0.00
PPIEP Costs	£1,534.00	£766.00	£2,300.00
Other Direct Costs	£19,897.00	£8,583.00	£28,480.00
Dissemination Costs	£0.00	£0.00	£0.00
Indirect Costs	£63,976.00	£29,218.00	£93,194.00
Total	**£139,385.00**	**£60,553.00**	**£199,938.00**

9. Management & Governance

Is Clinical Trials Authorisation required?	No

Does your project require ethics approval?	No

10. Uploads

References

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11. Administrative contact details

Administrative contact name	Christopher Bray

Administrative contact job title	Head of R&D Operations

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Administrative contact email address	christopher.bray@ouh.nhs.uk

12. Research & Development office contact

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R&D contact job title	Head of R&D Governance

R&D contact telephone number	0300 304 7777

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13. Acknowledgement, review and submit

COI declaration Please declare any conflicts or potential conflicts of interest that you or your co-applicants may have in undertaking this research, including any relevant personal, non-personal and commercial interest that could be perceived as a conflict of interest.
None

Lead Applicant - Agreement to terms and conditions

In **ticking** this, you as **Lead Applicant** confirm that the information given on this form is correct and that you will be actively engaged in this research and responsible for its overall management. In addition, you will accept responsibility for ensuring that the host institution and interested parties are kept informed. **Ticking this box constitutes an electronic signature of the Lead Applicant with regard to this application**
☒Confirmed

Appropriate and relevant involvement of patients/service users, carers and the public
☒Confirmed

A good quality plain English summary	☒Confirmed

A clear description of team member roles and contribution

☒Confirmed

A clear scientific abstract

☒Confirmed

A clear description of the changes from first stage

☒Confirmed

A flow diagram illustrating the study design / flow of participants (document upload), if appropriate

☒Confirmed

A full and accurate detailed budget breakdown

☒Confirmed

A clear justification of costs / value for money

☒Confirmed

References (document upload)

☒Confirmed

A clear Detailed Research Plan outlining the study design, methods, dissemination etc.

☒Confirmed

A CTU letter of support if required (document upload)

☒Confirmed

Completed Schedule of Events Cost Attribution Tool (SoECAT)

☒Confirmed

14. CV - Lead Applicant(s)

Full Name	Dr Karen Tiley	**Position**	Senior epidemiologist
Department	Paediatrics	**Institution**	University of Oxford
ORCID iD	0000-0003-2125-4125	**Telephone No.**	+44 07814527258
Address Line 1	Oxford Vaccine Group, CCVTM	**Country**	United Kingdom
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Address Line 3	Old Road, Headington	**Web Page.**	https://www.ovg.ox.ac.uk/team/karen-tiley

Qualification	Subject	Institution	From	To
PhD	Epidemiology of diphtheria in the UK and Europe	Manchester Metropolitan University	May 2007	May 2015
MSc	Epidemiology	Imperial College of Science, Technology and Medicine	September 2003	August 2004

MA	Biological natural Sciences	University of Cambridge	September 1999	July 2002

Selected recent relevant publication(s)

Publications
Tiley, KS., Ratcliffe, H., Voysey, M., Jefferies, K., Sinclair, G., Carr, M., Colin-Jones, R., Smith, D., Bowman, J., Hart, T., Kandasamy, R., Hinds, J., Gould, K., Berbers, G., Tcherniaeva, I., Robinson, H., Plested, E., Aley, P. and Snape, MD.. (2023) 'Nasopharyngeal Carriage of Pneumococcus in Children in England up to 10 Years After 13-Valent Pneumococcal Conjugate Vaccine Introduction: Persistence of Serotypes 3 and 19A and Emergence of 7C.', *The Journal of infectious diseases*, 227(5).
Ratcliffe, H., Tiley, KS., Longet, S., Tonry, C., Roarty, C., Watson, C., Amirthalingam, G., Vichos, I., Morey, E., Douglas, NL., Marinou, S., Plested, E., Aley, PK., Galiza, E., Faust, SN., Hughes, S., Murray, C., Roderick, MR., Shackley, F., Oddie, S., Lee, TWR., Turner, DPJ., Raman, M., Owens, S., Turner, PJ., Cockerill, H., Lopez Bernal, J., Ijaz, S., Poh, J., Shute, J., Linley, E., Borrow, R., Hoschler, K., Brown, KE., Carroll, MW., Klenerman, P., Dunachie, SJ., Ramsay, M., Voysey, M., Waterfield, T. and Snape, MD.. (2023) 'Serum HCoV-spike specific antibodies do not protect against subsequent SARS-CoV-2 infection in children and adolescents.', *iScience*, 26(12).
Ratcliffe, H., Tiley, KS., Andrews, N., Amirthalingam, G., Vichos, I., Morey, E., Douglas, NL., Marinou, S., Plested, E., Aley, P., Galiza, EP., Faust, SN., Hughes, S., Murray, CS., Roderick, M., Shackley, F., Oddie, SJ., Lees, T., Turner, DPJ., Raman, M., Owens, S., Turner, P., Cockerill, H., Lopez Bernal, J., Linley, E., Borrow, R., Brown, K., Ramsay, ME., Voysey, M. and Snape, MD.. (2022) 'Community seroprevalence of SARS-CoV-2 in children and adolescents in England, 2019-2021.', *Archives of disease in childhood*.
Wagner, KS., White, JM., Andrews, NJ., Borrow, R., Stanford, E., Newton, E. and Pebody, RG.. (2012) 'Immunity to tetanus and diphtheria in the UK in 2009.', *Vaccine*, 30(49).

Grants

Start Date	Duration	Amount	Source	Title	Role of Applicant

15. CV - Co-applicants

Co Applicant - 1

Full Name	Dr Dominic Kelly	**Organisation**	University of Oxford
Email Address	dominic.kelly@paediatrics.ox.ac.uk	**Web Page.**	

Education

From	To	Qualification	Subject	Country	Organisation	Department
06/2009	07/2009	CCT Paediatrics	General paediatrics (special interest in infection, immunology, allergy)	United Kingdom (England)	Royal College Paediatrics and Child Health	Post-graduate qualifications
10/2002	02/2008	PhD	Vaccine Immunology	United Kingdom (England)	The Open University	Post-graduate
06/1998	07/1998	MRCP (Paeds)	Paediatrics		Royal College of Physicians	Membership
10/1988	12/1993	MBBChir	Medicine	United Kingdom (England)	Cambridge University	Medical School

Publications

(2024) 'Carter, M., Shrestha, S., O'Reilly, P., Gurung, P., Gurung, M., Thorson, S., Kandasamy, R., Voysey, M., Kelly, S., Ansari, I., Shah, G., Amatya, P., Tcherniaeva, I., Berbers, G., Murdoch, DR, Pollard, AJ, Shrestha, S., & Kelly, DF Evaluation of acute and convalescent antibody concentration against pneumococcal capsular polysaccharides for the diagnosis of pneumococcal infection in children with community-acquired pneumonia. Paediatric Infectious Disease Journal (2024)',
(2023) 'Hillson K, Kantor J, Pinto MV, Pollard AJ, Kelly D, Vanderslott S. Motivations for paediatric vaccine trial participation. Trials. 2023;24(1).',
(2022) 'Bijker EM, Bateman EAL, Trück J, Patel S, Kelly DF. Screening for Immunodeficiencies in Children With Invasive Pneumococcal Disease: Six-year Experience From a UK Children's Hospital. Pediatric Infectious Disease Journal. 2022;41(7):575-8.',
(2022) 'Versteegen P, Barkoff AM, Valente Pinto M, van de Kasteele J, Knuutila A, Bibi S, de Rond L, Teräsjärvi J, Sanders K, de Zeeuw-Brouwer ML, Luoto R, ten Hulscher H, Clutterbuck EA, Sanders EAM, Mertsola J, Berbers GAM, He Q, Kelly DF, Buisman AM. Memory B Cell Activation Induced by Pertussis Booster Vaccination in Four Age Groups of Three Countries. Frontiers in Immunology. 2022;13.',
(2021) 'Versteegen P, Valente Pinto M, Barkoff AM, van Gageldonk PGM, Kassteele DJVD, van Houten DMA, Sanders PEAM, de Groot PR, Diavatopoulos DDA, Bibi DS, Luoto DR, He PQ, Buisman DAM, Kelly DF, Mertsola PJ, Berbers DGAM. Responses to an acellular pertussis booster vaccination in children, adolescents, and young and older adults: A collaborative study in Finland, the Netherlands, and the United Kingdom. EBioMedicine. 2021;65.',
(2022) 'Shrestha S, Gurung M, Amatya P, Bijukchhe S, Bose AS, Carter MJ, Gautam MC, Gurung S, Hinds J, Kandasamy R, Kelly S, Khadka B, Maskey P, Mujadidi YF, O'Reilly PJ, Pokhrel B, Pradhan R, Shah GP, Shrestha S, Wahl B, O'Brien KL, Knoll MD, Murdoch DR, Kelly DF, et al. Effect of the of 10-valent pneumococcal conjugate vaccine in Nepal 4 years after introduction: an observational cohort study. The Lancet Global Health. 2022;10(10):e1494-e504.',

Grants

Start Date	Duration	Amount	Source	Title	Role of Applicant
01/03/2024	60	379,419.00	NIHR	NIHR153577 The Monoclonal Antibody Medications in inflammatory Arthritis: stopping or continuing in pregnancy (MAMA) trial	PI for immunology studies (funding quoted)
01/06/2022	18	102,747.00	Iliad Biotech	Phase 2B study of BPZE1 in children	Oxford PI
01/09/2018	48	127,337.00	MRC and Kymab (iCASE studentship)	Computational methods for rapid structural modelling of antigen–antibody interactions to improve identification of antigen-specific antibodies from Ig-seq repertoire data	Co-investigator
01/03/2016	60	1,200,000.00	Bill and Melinda Gate and IMI (EU)	: Pertussis Correlates of Protection in Europe – clinical trials of pertussis vaccines in infants, children and adults	PI for Oxford (funding amount quoted)
01/01/2014	48	1,300,000.00	GAVI	Evaluation of pneumococcal vaccine effectiveness and impact in urban Nepal. Co-investigato	Co-applicant

Grants

Start Date	Duration	Amount	Source	Title	Role of Applicant
01/01/2013	48	3,500.00	ESPID	Study of use of pneumococcal conjugate vaccine (Prevenar 13) in children aged 6 and over with type 1 diabetes mellitus	Principal applicant

Co Applicant - 2

Full Name Ms Sophia Wilkinson **Organisation** University of Oxford
Email Address sophia.wilkinson@paediatrics.ox.ac.uk **Web Page.**

Education

From	To	Qualification	Subject	Country	Organisation	Department
09/1999	09/2000	MSc(Hons)	Development Practices (Humanitarianism)	United Kingdom	Oxford Brookes University	School of Architecture
09/1994	07/1995	Postgraduate Diploma	Radio Journalism	United Kingdom	University of the Arts, London	London College of Communication
09/1991	07/1994	BA(Hons)	Thai Studies and History	United Kingdom	School of Oriental and African Studies	Thai Studies

Publications

Grants

Start Date	Duration	Amount	Source	Title	Role of Applicant